

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
5	(a)		$\lambda = \frac{\ln 2}{T_{\frac{1}{2}}} = \frac{\ln 2}{3.8} \text{ day}^{-1} [=0.182 \text{ day}^{-1}] \text{ (1)}$ Activity after 12 days $A = A_0 \exp\left(-\frac{12 \ln 2}{3.8}\right)$ (1) substitution $= 0.112A_0$ (1) $\therefore$ % reduction = 88.8%(1) (Accept variation because of rounding) Alternative Number of half-lives, $n = \frac{12}{3.8} = 3.16$ [or by implication] (1) Fraction after 12 days $= 2^{-3.16}$ (1) = 0.112 (1) = 11.2% $\therefore$ Percentage reduction = 88.8% (1)		4		4	4	